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In Bayes' Theorem View: Infection Test Outcomes Analysis

1 Background

Have you ever wondered how the mathematics you learn in A-Level classes lays the foundation for cutting-edge university research? **Bayes' Theorem** serves as a perfect bridge. This poster brings it to life through a scenario from the **COVID pandemic**: calculating the true probability of infection following a Lateral Flow Test and a Confirmation Test (PCR test). Given that the probability of each elementary event is known for all testing stages, how can we determine the actual probability of infection when diagnosed with the confirmation test?

2 Introduction

E^c is preferred for symbolizing the **complement** of the event E (in Probability and Statistics module of university).

The real-valued number set function P is defined as a probability measured on the sample space Ω.

$P(E|F) = \frac{P(E \cap F)}{P(F)}$: the probability of an event E **given** an event F.

Sensitivity: The possibility of identifying a person infected with the disease correctly.

Specificity: The possibility of identifying a person not infected with the disease correctly.

(University of Nottingham, 2025)

3 Derivation

Theorem of Total Probability

$E_i \cap E_j = \emptyset$

$E_1 \cup E_2 \cup \cdots \cup E_n = \Omega$

$F \subseteq \Omega, F = (F \cap E_1) \cup (F \cap E_2) \cup \cdots \cup (F \cap E_n)$

$$P(F) = P(F \cap E_1) + P(F \cap E_2) + \cdots + P(F \cap E_n) = \sum_{i=1}^{n} P(F|E_i)P(E_i)$$

$$P(E_i|F) = \frac{P(E_i \cap F)}{P(F)} = \frac{P(F|E_i)P(E_i)}{P(F)}$$

$$P(E_i|F) = \frac{P(F|E_i)P(E_i)}{\sum_{j=1}^{n} P(F|E_j)P(E_j)}$$

Bayes' Theorem

4 Methods

1. Known Data and Corresponding Expressions:

1.1 **Basic elementary events**:

D-Event person is infected with COVID-19 (D^c- not infected).

1.2 **Sensitivity** and **Specificity** for two tests:

-The Lateral Flow test: $P(L|D) = 0.85$-Sensitivity, $P(L^c|D^c) = 0.975$-Specificity.

-PCR test: $P(R|D) = 0.98$-Sensitivity, $P(R^c|D^c) = 0.995$-Specificity.

-The probability of being **infected** with COVID-19 is p.

2. Substitute the data:

Now we apply Bayes' Theorem, $P(D|L,R) = \frac{P(L,R|D) \times P(D)}{P(L,R)}$.

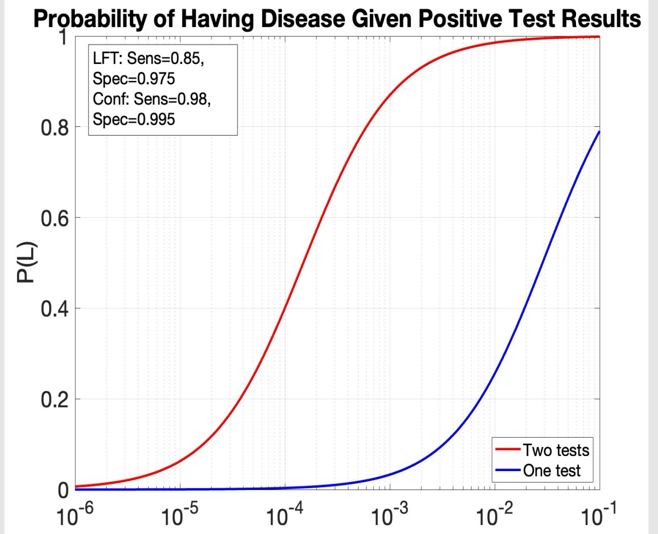
With **Conditional Probability**:

Given a person's true health status (whether infected or not),

the results of the two tests are **independent**.

This assumption allows us to factor the joint probability:

$P(L \cap R|D) = P(L|D) \times P(R|D)$, $P(L \cap R|D^c) = P(L|D^c) \times P(R|D^c)$.



Probability of truly having COVID-19 given two positive tests with p ranging from 10^{-6} to 10^{-1}

With **Theorem of Total Probability**:

We regard E as D (or D^c), F as $L \cap R$ in the formula in Derivative, we can draw: $P(L,R) = P(L,R|D)P(D) + P(L,R|D^c)P(D^c)$.

Again, with conditional probability, it can be expressed as:

$$P(L|D) \cdot P(R|D) \cdot P(D) + P(L|D^c) \cdot P(R|D^c) \cdot P(D^c).$$

From above, the final formula is:

$$P(D|L,R) = \frac{P(L|D) \cdot P(R|D) \cdot P(D)}{P(L|D) \cdot P(R|D) \cdot P(D) + P(L|D^c) \cdot P(R|D^c) \cdot P(D^c)}$$

We can **substitute the known data** (also did in previous steps):

$$P(D|L,R) = \frac{0.833p}{0.833p + 0.000125(1-p)}$$

5 Application

Bayesian methods combine prior knowledge with new data to enhance decision-making across various fields. In **machine learning**, they improve predictions through continuous data integration. For **medical diagnosis**, they combine disease prevalence with patient symptoms to aid decision making.

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However, these applications still face **computational demands** and ethical concerns regarding data privacy and **fairness**. These challenges require careful application and ongoing improvements (Song, 2025).

6 Conclusion

According to our above analysis, low-prevalence populations, even highly confirmed positive test results may point to "false positives". More realistic and sensible results of probability judgments reveal the power of Bayes' theorem, combining test results with prior knowledge.

This breaks the intuitive notion of "test as truth" and demonstrates the profound value of probabilistic thinking in dealing with uncertainties in the real world.

Reference

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